

The ECD completion of Hermitian maps is generated by its CP cone

Candidate full proof of the conjecture in arXiv:1810.10922

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Abstract

Let G be a positive discrete unbounded operator. Shirokov conjectured that every Hermitian-preserving map in the energy-constrained-diamond (ECD) completion $\mathfrak{Y}_G(A, B)$ is a difference of two maps in the completed completely-positive cone $\mathfrak{F}_G^0(A, B)$. We prove the conjecture. Fixing $E_0 > 0$ and writing $R = (I + G/E_0)^{-1/2}$, the weighted map $\widehat{\Phi}(\rho) = \Phi(R\rho R)$ is an ordinary completely bounded map, and its diamond norm is equivalent to the ECD norm, with constants 1 and 2. Discreteness of G makes R compact. Hence the normal dual of $\widehat{\Phi}$ is a cb-norm limit of normal Hermitian maps taking values in finite spectral corners. A summable telescoping approximation, normal Wittstock decomposition in each corner, and conjugation by the bounded finite-corner inverse of R produce two ECD-convergent series of bounded CP maps. Their limits lie in $\mathfrak{F}_G^0(A, B)$ and their difference is Φ .

1 Source problem

The source is M. E. Shirokov, *On completion of the cone of CP linear maps with respect to the energy-constrained diamond norm*, arXiv:1810.10922v2 (2019) [1]. On page 23 it asks:

It is well known that any map Φ in $\mathfrak{Y}(A, B)$ can be represented as $\Psi_1 - \Psi_2$, where Ψ_1 and Ψ_2 are maps in $\mathfrak{F}(A, B)$ [13, 26]. Theorem 3B gives a reason for the following

Conjecture. Any map Φ in $\mathfrak{Y}_G(A, B)$, where G is a positive discrete unbounded operator, can be represented as $\Phi = \Psi_1 - \Psi_2$, where Ψ_1 and Ψ_2 are maps in $\mathfrak{F}_G^0(A, B)$ [22]

Theorem 3B implies the following characterisation of the class of maps in $\mathfrak{Y}_G(A, B)$ which have such representation.

In the source notation, $\mathfrak{T}_G(\mathcal{H}_A)$ is the linear span of positive trace-class operators of finite G -energy. The real Banach space $\mathfrak{Y}_G(A, B)$ consists of Hermitian-preserving maps on this domain whose amplifications have the required continuity on every subnormalized energy-bounded positive set. Its norm is

$$\|\Phi\|_{\diamond, E}^G = \sup\{\|(\Phi \otimes \text{Id}_R)(\omega)\|_1 : \omega \in \mathfrak{S}(\mathcal{H}_A \otimes \mathcal{H}_R), \text{Tr } G\omega_A \leq E\}.$$

For discrete G , Theorem 3B of the source identifies $\mathfrak{Y}_G(A, B)$ with the ECD completion of the ordinary Hermitian completely bounded maps $\mathfrak{Y}(A, B)$. Its Theorem 1B identifies $\mathfrak{F}_G^0(A, B)$ with the ECD completion of the cone $\mathfrak{F}(A, B)$ of ordinary bounded CP maps.

Here “discrete” has the source’s meaning: G has finite-multiplicity eigenvalues tending to $+\infty$. A zero ground-state energy is allowed.

2 Proof intuition

The ECD norm looks weaker than the diamond norm only because high-energy directions are discounted. The operator

$$R = (I + G/E_0)^{-1/2}$$

builds that discount into the input. Passing from Φ to $\widehat{\Phi}(\rho) = \Phi(R\rho R)$ turns the ECD problem into an ordinary diamond-norm problem, with no loss beyond a factor 2. Discreteness of G is used exactly once: it makes R compact, so the dual of every weighted map can be approximated by finite spectral corners.

Finite-corner Hermitian cb maps admit normal Wittstock decompositions into differences of CP maps. Decomposing a naively chosen approximating sequence would not suffice, because the positive and negative parts could diverge. The key bookkeeping step is instead to choose a summably accurate sequence of corners and decompose its telescoping increments. The CP norms are then summable. On a finite spectral corner R has a bounded inverse, so each CP piece can be lifted back to an ordinary bounded CP map before taking its ECD limit.

3 The weighted norm equivalence

Fix $E_0 > 0$ and set

$$S = I + G/E_0, \quad R = S^{-1/2}.$$

Thus $0 < R \leq I$, $\text{Ran } R = \mathcal{D}(\sqrt{G})$, and $GR^2 \leq E_0 I$.

Lemma 1 (Weighted transform). *For every $\Phi \in \mathfrak{Y}_G(A, B)$, the formula*

$$\widehat{\Phi}(\rho) = \Phi(R\rho R), \quad \rho \in \mathfrak{T}(\mathcal{H}_A),$$

defines a Hermitian completely bounded map $\widehat{\Phi} : \mathfrak{T}(\mathcal{H}_A) \rightarrow \mathfrak{T}(\mathcal{H}_B)$, and

$$\|\widehat{\Phi}\|_{\diamond} \leq \|\Phi\|_{\diamond, E_0}^G \leq 2\|\widehat{\Phi}\|_{\diamond}. \quad (1)$$

The construction commutes with amplification and is injective on $\mathfrak{Y}_G(A, B)$.

Proof. Let ω be a state on A together with an arbitrary ancilla, and put

$$\sigma = (R \otimes I)\omega(R \otimes I).$$

Then $\sigma \geq 0$, $\text{Tr } \sigma \leq 1$, and

$$\text{Tr } G\sigma_A = \text{Tr}(GR^2)\omega_A \leq E_0.$$

The subnormalized-state estimate in the proof of Theorem 3A of the source therefore gives

$$\|(\widehat{\Phi} \otimes \text{Id})(\omega)\|_1 \leq \|\Phi\|_{\diamond, E_0}^G.$$

For Hermitian maps the diamond norm is the supremum over states. This proves that $\widehat{\Phi}$ extends to a completely bounded map and proves the first inequality in (1).

Conversely, let ω be a state with $\text{Tr } G\omega_A \leq E_0$. The positive quadratic-form expression

$$\tau = (S^{1/2} \otimes I)\omega(S^{1/2} \otimes I)$$

is trace class and satisfies

$$\text{Tr } \tau = 1 + E_0^{-1} \text{Tr } G\omega_A \leq 2, \quad (R \otimes I)\tau(R \otimes I) = \omega.$$

Consequently

$$\|(\Phi \otimes \text{Id})(\omega)\|_1 = \|(\hat{\Phi} \otimes \text{Id})(\tau)\|_1 \leq 2\|\hat{\Phi}\|_\diamond.$$

Taking the supremum proves the second inequality.

For injectivity, suppose $\hat{\Delta} = 0$. If $x \in \mathcal{D}(\sqrt{G}) = \text{Ran } R$, then $|x\rangle\langle x| = R|a\rangle\langle a|R$ for some a , so $\Delta(|x\rangle\langle x|) = 0$. Every finite-energy positive trace-class operator has a rank-one decomposition whose vectors lie in $\mathcal{D}(\sqrt{G})$ and whose finite partial sums remain in a common subnormalized energy-bounded set. Those sums converge in trace norm, and the defining continuity of $\Delta \in \mathfrak{Y}_G$ gives $\Delta(\rho) = 0$. Linearity on the span $\mathfrak{T}_G(\mathcal{H}_A)$ proves $\Delta = 0$. $\square$

4 Full solution

Theorem 2. *Let G be a positive discrete unbounded operator on $\mathcal{H}_A$. For every $\Phi \in \mathfrak{Y}_G(A, B)$ there are $\Psi_+, \Psi_- \in \mathfrak{F}_G^0(A, B)$ such that*

$$\Phi = \Psi_+ - \Psi_-.$$

Thus the conjecture on page 23 of arXiv:1810.10922 is true.

Proof. Let $T = \hat{\Phi}^* : \mathfrak{B}(\mathcal{H}_B) \rightarrow \mathfrak{B}(\mathcal{H}_A)$. This is a normal Hermitian completely bounded map and $\|T\|_{\text{cb}} = \|\hat{\Phi}\|_\diamond$.

By Theorem 3B of the source, choose $\Phi_m \in \mathfrak{Y}(A, B)$ converging to Φ in ECD norm. Lemma 1 shows that $\hat{\Phi}_m \rightarrow \hat{\Phi}$ in diamond norm. On duals,

$$T_m(X) = \hat{\Phi}_m^*(X) = R\Phi_m^*(X)R \longrightarrow T(X) \quad \text{in cb norm.} \quad (3)$$

Let P_N be the spectral projection onto the first N eigenvectors of G , counted with multiplicity, and put $R_N = P_N R = R P_N$. Since G is discrete, R is compact and $\|R - R_N\| \rightarrow 0$. For fixed m ,

$$T_{m,N}(X) = P_N T_m(X) P_N = R_N \Phi_m^*(X) R_N$$

is a normal Hermitian cb map with values in the finite corner $P_N \mathfrak{B}(\mathcal{H}_A) P_N$, and

$$\|T_m - T_{m,N}\|_{\text{cb}} \leq 2\|R\| \|\Phi_m\|_\diamond \|R - R_N\| \longrightarrow 0. \quad (4)$$

Equations (3)–(4) show that T is in the cb-norm closure of the normal Hermitian finite-corner maps.

Choose such maps S_k , supported on an increasing sequence of spectral corners, so that

$$\|T - S_k\|_{\text{cb}} \leq 4^{-k}.$$

Set $S_0 = 0$ and $D_k = S_k - S_{k-1}$. Then

$$T = \sum_{k=1}^{\infty} D_k \quad \text{in cb norm,} \quad \sum_{k=1}^{\infty} \|D_k\|_{\text{cb}} < \infty. \quad (2)$$

Each D_k is normal, Hermitian, and supported in some finite spectral corner $P_{N_k} \mathfrak{B}(\mathcal{H}_A) P_{N_k}$.

The normal Wittstock decomposition theorem [2, 3] gives normal CP maps $C_{k,+}, C_{k,-}$ into that same corner such that

$$D_k = C_{k,+} - C_{k,-}, \quad \|C_{k,\pm}\|_{\text{cb}} \leq \|D_k\|_{\text{cb}} + 2^{-k}. \quad (5)$$

For completeness, the norm control follows from a normal factorization $D_k(X) = \frac{1}{2}[U^*\pi(X)V + V^*\pi(X)U]$, after balancing U, V , and the identity

$$D_k(X) = \frac{1}{4}(U + V)^*\pi(X)(U + V) - \frac{1}{4}(U - V)^*\pi(X)(U - V).$$

An arbitrarily small factorization error yields the 2^{-k} allowance. By (2), both series of CP cb norms are summable.

On $P_{N_k}\mathcal{H}_A$, the inverse $R_{N_k}^{-1} = P_{N_k}R^{-1}P_{N_k}$ is bounded. Define a normal CP map

$$L_{k,\pm}(X) = R_{N_k}^{-1}C_{k,\pm}(X)R_{N_k}^{-1}, \quad X \in \mathfrak{B}(\mathcal{H}_B),$$

and let $\psi_{k,\pm} : \mathfrak{T}(\mathcal{H}_A) \rightarrow \mathfrak{T}(\mathcal{H}_B)$ be its predual. Thus $\psi_{k,\pm}$ is an ordinary bounded CP map. Its weighted dual is exactly the desired corner piece:

$$\widehat{\psi}_{k,\pm}^*(X) = RL_{k,\pm}(X)R = C_{k,\pm}(X). \quad (6)$$

Lemma 1, (5), and (6) give

$$\sum_k \|\psi_{k,\pm}\|_{\diamond, E_0}^G \leq 2 \sum_k \|C_{k,\pm}\|_{\text{cb}} < \infty. \quad (7)$$

The partial sums $\sum_{k \leq n} \psi_{k,\pm}$ are ordinary bounded CP maps. By (7) they converge in ECD norm, and Theorem 1B of the source says that their limits $\Psi_{\pm}$ belong to $\mathfrak{F}_G^0(A, B)$. Taking weighted duals and using (2) and (6) yields

$$\widehat{\Phi}^* = T = \sum_k (C_{k,+} - C_{k,-}) = \widehat{\Psi}_+^* - \widehat{\Psi}_-^*.$$

Hence $\widehat{\Phi} = \widehat{\Psi}_+ - \widehat{\Psi}_-$. Injectivity in Lemma 1 finally gives $\Phi = \Psi_+ - \Psi_-$ on $\mathfrak{T}_G(\mathcal{H}_A)$. $\square$

5 Checks, scope, and novelty guardrail

The factor 2 in Lemma 1 comes exactly from $\text{Tr}[(I + G/E_0)\omega_A] \leq 2$; no finite-dimensional approximation is hidden in that step. The only use of discreteness is compactness of R in (4). The proof does not claim the conjecture for a positive G with noncompact resolvent.

The telescoping construction is essential: it turns convergence of Hermitian corners into absolute summability of their CP parts. Normality is also essential because it ensures that every lifted CP map has a predual on trace class. The finite-corner inverse is always bounded, even though R^{-1} is unbounded on the full Hilbert space.

On 11 August 2026, bounded novelty searches covered the run’s registry, solution, attempt, and proof-gap indexes; the exact source identifier and notation $\mathfrak{Y}_G(A, B)$ and $\mathfrak{F}_G^0(A, B)$; and arXiv/web-facing variants of “ECD completion difference of completely positive maps”, “Hermitian energy-constrained diamond norm Wittstock”, and the exact conjecture text. They found the source, general completely bounded/decomposable-map literature, and an already-recorded answer to the source’s separate operator E -norm question, but no later resolution of this conjecture or this weighted compact-corner argument. This is a bounded novelty guardrail, not a claim of priority.

Human-review recommendation. Review as a candidate full solution. The three highest-value checks are: (i) the quadratic-form identity used for τ in Lemma 1; (ii) the normal, norm-controlled Wittstock decomposition in finite corners; and (iii) the passage from vanishing on weighted rank-one operators to all of $\mathfrak{T}_G(\mathcal{H}_A)$ in the injectivity argument.

References

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